



Design and Prototyping Exercise:

Option#2 - DeskDash

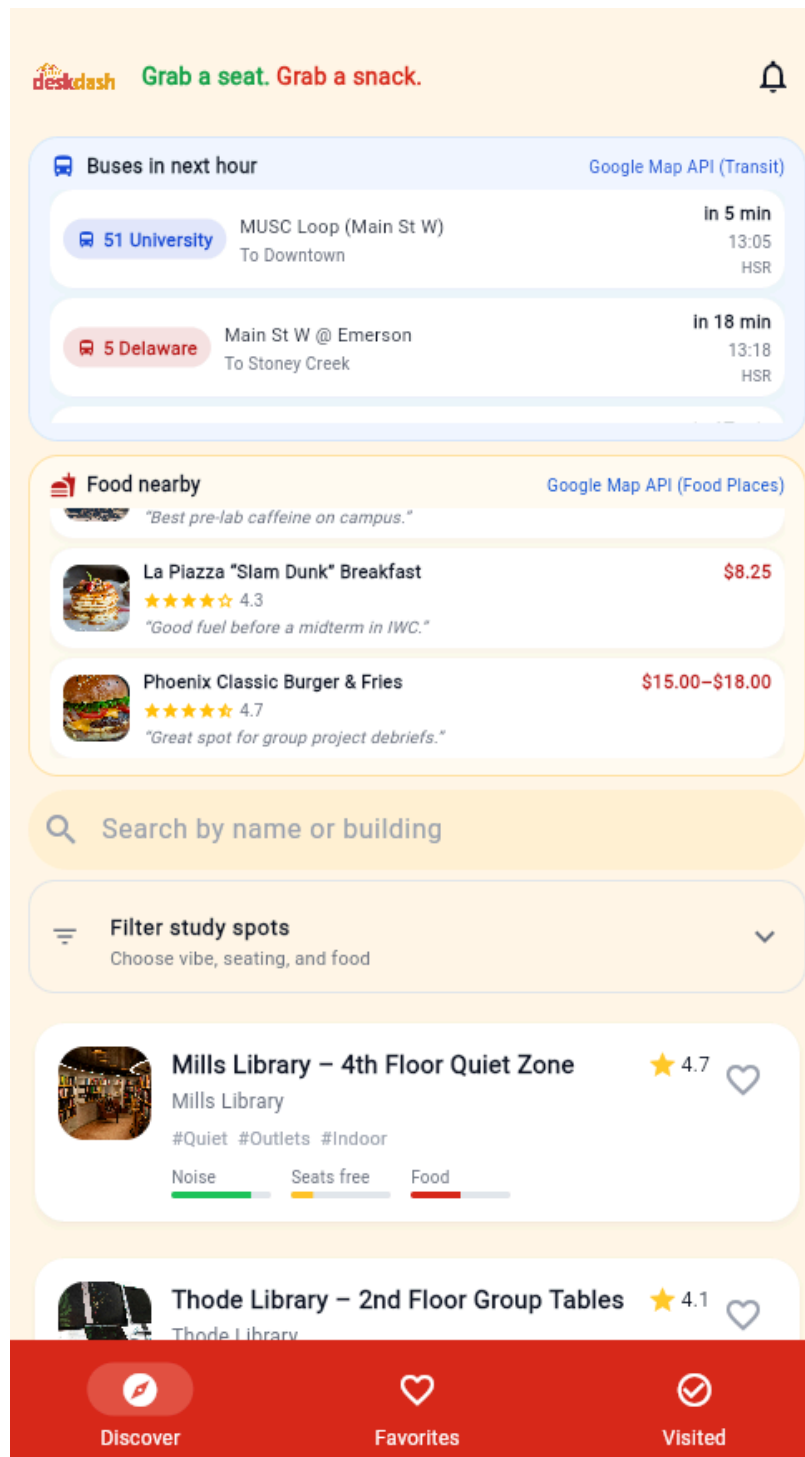
Course: SFWRENG 4HC3/6HC3 (Fall 2025)

Developer: Zifan Si

App Github Repo: <https://github.com/ZifanSi/DeskDash>

PLEASE CHECK README FOR MOBILE APP SET UP !

Section A - Summary of User Characteristics and Needs



1.1. Target User Groups

User Group #	
Primary	Current undergraduate students at McMaster
Secondary	Campus members who are not undergrads
Tertiary	Visitors and short-term guests

A1. Undergraduate students at McMaster (primary group): The main users are McMaster undergrads in programs like Engineering, Health Sci, Science, and Business. They move between buildings such as ITB, JHE, TSH, L.R. Wilson, and MUSC all day, usually with only a few minutes between classes or labs. They pull out their phone in the hallway and quickly check DeskDash to find a nearby spot to sit, instead of wandering around. For example, an Engineering student leaving ITB might use DeskDash to see if there are any quieter seats left in Thode or Mills before their next lecture.

- DeskDash is designed mainly for them, so all core features (listings, filters, reviews, favorites, bus, food) target their day-to-day study needs.

A2. Graduate students, TAs, and research assistants (secondary group): Graduate students, TAs, and research assistants also spend long days on campus, often split between offices, labs, and classrooms. **Even if they have an office**, they still need quiet spaces to mark, prepare tutorials, or read papers. They prefer study spots close to their home buildings (e.g., ITB, ABB, HSC). With DeskDash on their phone, they can quickly check noise and crowd levels before heading to a library floor or common area, which is especially helpful during midterms or exam periods when campus is busy.

- They share many of the same needs (Food, shortest path of locations b/w meetings, Bus when ready to leave office at late night), so they can still use DeskDash, even though they are not the main design focus

A3. Campus visitors and short-term guests (tertiary group): This group includes people who are not full-time McMaster students but still spend a few hours on campus and need somewhere to sit and work. Examples are prospective students and parents on campus tours, alumni visiting for events, conference attendees in buildings like L.R. Wilson, or partners visiting labs for meetings. They often do not know the campus layout or which buildings are good for studying. With DeskDash on their phone, they can quickly see a list of nearby study spaces, basic info like “Quiet / Group / Near food / Bus” and simple reviews. This lets them find a decent place to sit with Wi-Fi, maybe near MUSC or Mills, without having to ask for directions or guess which floor of a library is suitable.

- They are not the priority, but DeskDash should still be understandable without insider knowledge, so that a visitor can open the app and quickly find a spot.

1.2 Key User Characteristics

Mobile, on-the-move usage

Most students open DeskDash on their phone while walking between ITB, JHE, Mills, Thode, or MUSC, or while waiting in line for coffee. They often use one hand and their thumb, so the app has to work in quick, small interactions on a small screen. This is why the home screen puts everything in one vertical scroll: study cards, Food nearby, and the bus widget.

Short, time-boxed study and travel windows

Many students only have 10 to 30 minutes between lectures, labs, or a bus/GO trip. They do not have time to wander around Mills to check every floor or to look up bus times on a separate app. DeskDash supports this by showing live-style “Seats free” bars on each card and a compact bus widget with the next few departures, so they can quickly decide: “Do I sit here, or do I pack up for the next bus?”

Campus mental model based on buildings and routes

Students think in buildings (“Mills”, “Thode”, “MUSC”, “Wilson”) and bus routes (like the 51 or 5 around campus), not street addresses. They choose where to study based on where they are now and where they need to go next. DeskDash follows this mental model by listing study places by building and tying extra widgets (Food nearby and bus times) to the student’s current campus location.

Linked habits: study, food, and going home

Long study blocks are often planned around nearby food and the time of the next bus home. A student might think: “Study in Mills until I’m hungry, grab something at La Piazza, then catch the next bus.” DeskDash reflects this pattern with three supporting pieces: a Food nearby panel, a “Food” bar on each study card, and a small bus widget that surfaces buses in the next hour. Together with the “Seats free” and “Noise” bars, this helps students make one quick decision that fits studying, eating, and getting home on time.

1.3 Core Needs and Challenges

Need 1: Find an open seat fast

McMaster students often walk into Mills, Thode, or MUSC and see whole areas packed, especially between classes. They don’t have time to wander floor-to-floor. DeskDash cards and the “Live comfort snapshot” use Seats free bars and simple labels (e.g., 40 / 180) so a student can quickly spot which spaces are likely to have a seat right now.

Need 2: Pick a study vibe that matches the task

Sometimes they need silent solo work (exam prep in Mills), other times they want a louder table for group work (Thode, MUSC). It's hard to guess this just from the building name. DeskDash shows Noise and Crowd bars and short reviews ("usually silent in the mornings") so students can choose a spot that fits their current task instead of gambling.

Need 3: Plan study time around nearby food

Long study blocks usually happen near food: La Piazza, E-Café, Phoenix, Tim's, etc. Students want to know if they can grab something quick without leaving the area. The Food nearby panel in DeskDash shows a tiny "menu" of nearby options with rating, quote, and price range, plus a "Food" bar on each study card, so they can see at a glance which study spots are close to decent food.

Need 4: Match study time with bus / transit schedule

Many McMaster students watch HSR routes like 51 University or 5 Delaware to get home. They often ask "Do I have 15–20 minutes to study before my bus comes?" The Buses in next hour widget in DeskDash shows the next few departures (route, stop, destination, minutes) so they can pick a nearby study spot and use that dead time well, instead of just waiting at the loop or missing a bus.

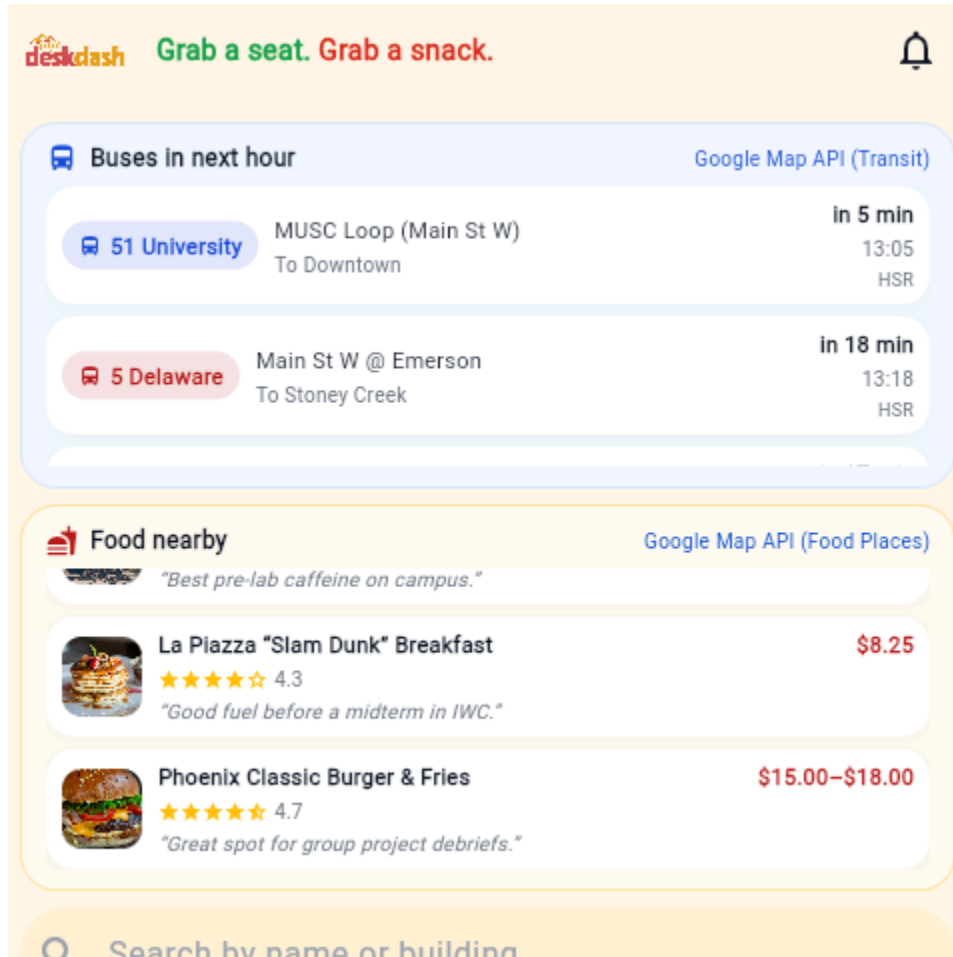
Challenge 1: Campus information is scattered across many places

Right now, students have to use multiple tools: McMaster website for library hours, Google Maps for food, HSR/Transit apps for buses, and their own memory for "good" study spots. This makes simple decisions ("Where do I sit for 40 minutes near MUSC before the 51 comes?") much slower. DeskDash tries to pull the most important parts into one mobile view: study cards, comfort snapshot, Food nearby, and Buses in next hour, so students don't have to switch apps while they are walking.

Challenge 2: Decisions have to be made in under 30 seconds

Most of these choices happen in short gaps: stepping out of a lecture, walking through MUSC, or leaving Mills before a bus. Students don't have time to read long descriptions or click through deep menus. If the interface is slow or cluttered, they will just go back to the place they already know, even if it's full or loud. DeskDash is designed so key signals (Noise, Seats free, Food, next buses) are visible directly on the main screen and on the first scroll, helping them make a "good enough" decision in a few seconds.

Section B - Additional Task/Function



3.1 Food nearby (Location API idea)

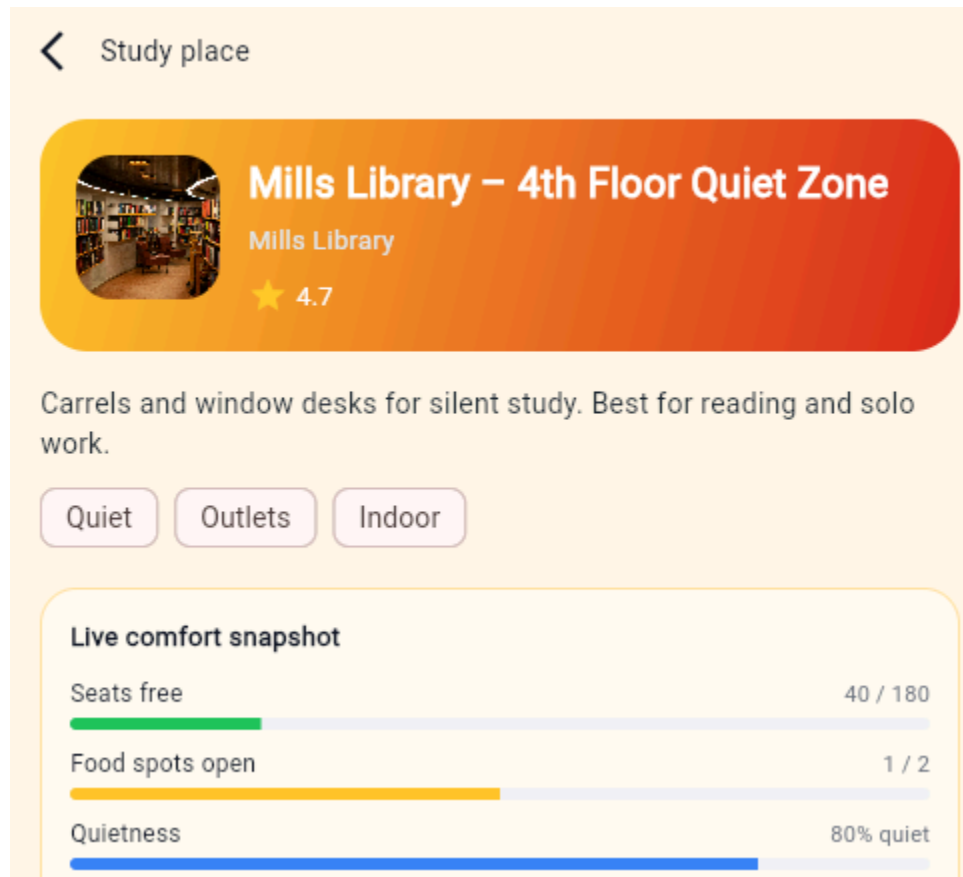
The Food nearby panel on the home screen shows a short, scrollable list of nearby food options such as E-Café, La Piazza, or Phoenix. Each row has a picture, name, rating, quote (e.g., “Best pre-lab caffeine on campus.”), and a price range. Tapping a row opens the restaurant/menu link in the browser. In a full version, the “Google Map API (Food Places)” label and the blue “Location API” text indicate that the app would use the student’s current location on campus to pull the closest food places around Mills, MUSC, ETB, etc. This supports the need to plan study time around nearby food—students can quickly see where they can sit *and* where they can grab something without leaving the area.

3.2 Buses in next hour (Transit helper)

Above the food panel, the Buses in next hour widget shows upcoming HSR routes that McMaster students actually use, such as 51 University and 5 Delaware. For each trip, the card lists the route, stop (e.g., MUSC Loop), destination (e.g., “To Downtown”), departure time, and “in X min” label.

This feature is tied to the need to match study time with bus / transit schedules. Commuter students can quickly see whether they have 5, 15, or 25 minutes before the next bus, and decide if it's worth sitting down in Mills, Thode, or MUSC instead of just standing at the loop. It turns "dead time" into usable study time.

3.3 Live comfort snapshot (seats, food, noise)



On each study place details screen, the Live comfort snapshot (mock) card combines three pieces of capacity-style information:

1. Seats free – for example, 40 / 180, with a green bar showing how much seating is left.
2. Food spots open – for example, 1 / 2, showing how many nearby food options are currently open.
3. Quietness – a blue progress bar such as 80% quiet, based on a simple noise score.

In this prototype the numbers are mock data, but they demonstrate how a real system could show real-time comfort based on seats, food, and noise together. This directly supports the need to find an open seat fast and pick the right vibe for the task—students don't just see "Mills Library," they see whether it's likely to be usable right now.

Section C - Table/Diagram of All Functions

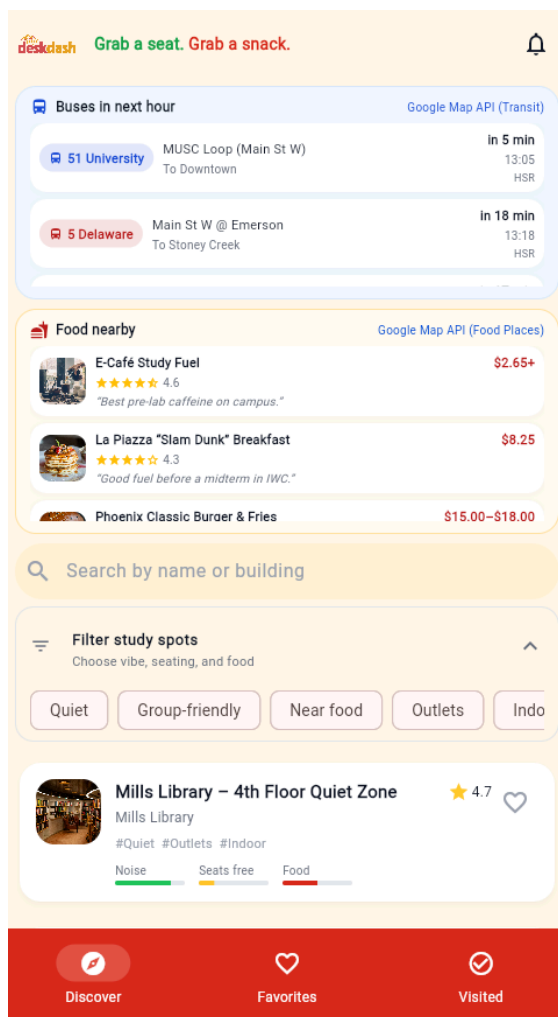
#	Task / Function	What the user can do	Key interface screen(s) / widget(s)
#1	Browse study places	Scroll a vertical feed of study-place cards showing name, building, rating, tags, and mini Noise / Seats free / Food bars.	Home screen – Discover tab (ListView of cards)
#2	Search by name or building	Enter "Mills", "Thode", "MUSC", etc. to live-filter the list of cards.	Home screen – search TextField ("Search by name or building")
#3	Filter by study needs	Toggle FilterChips like Quiet, Group-friendly, Near food, Outlets, Indoor, Outdoor to refine results.	Home screen – filter chips row
#4	View full study place details	Tap a card to push a details screen with title, building, description, key tags (Quiet / Outlets / Indoor), rating, and reviews.	Place details screen (navigated from Home card tap)
#5	See "Live comfort snapshot" (additional)	Read a compact status block with progress bars for Seats free, Food spots open, and Quietness.	Place details screen – "Live comfort snapshot (mock)" container
#6	Mark / unmark as Favorite	Tap the heart to add/remove a place from the user's favourites list.	Home screen – heart IconButton on card; Place details screen – "Save to favorites" FilledButton; Favorites tab (ListView of favourite places)
#7	Mark / unmark as Visited	Mark that the user has already tried a place, or undo it.	Place details screen – "Mark visited" OutlinedButton; Home screen – "Visited" status row; Visited tab (ListView of visited places)

#	Task / Function	What the user can do	Key interface screen(s) / widget(s)
#8	Read existing reviews	Scan student-style review chips/cards about noise, outlets, and general vibe.	Place details screen – Reviews section (Column of _ReviewCards)
#9	Write and post a review	Tap "Write a review", type into a multi-line field, and post so the review appears at the top of the list.	Place details screen – "Write a review" TextButton.icon → Write review bottom sheet (showModalBottomSheet)
#10	View food options nearby (additional)	See a compact list of nearby food items with image, name, rating, quote, and price range, driven by the Location API idea (current location → closest food).	Home screen – "Food nearby" panel (header row + scrolling ListView of food items)
#11	Open restaurant / menu link	Tap a food row to launch the restaurant/menu URL in the external browser.	Home screen – "Food nearby" panel (row InkWell → launchUrl)
#12	Switch between Discover, Favorites, and Visited	Change the primary content to all places, favourites only, or visited only.	Bottom NavigationBar (tabs: Discover, Favorites, and Visited)
#13	View buses in next hour	Check upcoming HSR departures (e.g., 51 University, 5 Delaware) with route, stop, destination, departure time, and "in X min" label so students can match short study sessions with transit.	Home screen – "Buses in next hour" panel (header row + list of bus cards)

Section D - Design Justifications (Norman's principles)

4.1 Visibility

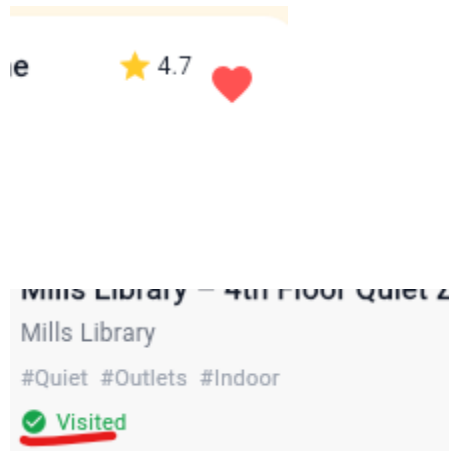
Visibility means users can easily see what they can do next. On the DeskDash home screen, key actions are visible in one short scroll: the search bar and filter chips (Quiet, Group-friendly, Near food, Outlets, Indoor), the Buses in next hour panel, the Food nearby panel, the vertical list of study cards (with Noise / Seats free / Food mini bars), and the bottom tabs (Discover, Favorites, Visited). On the place details screen, the Live comfort snapshot, Save to favorites, Mark visited, and Reviews / Write a review are all clearly labeled and grouped, so students don't have to guess where features are.



4.2 Feedback

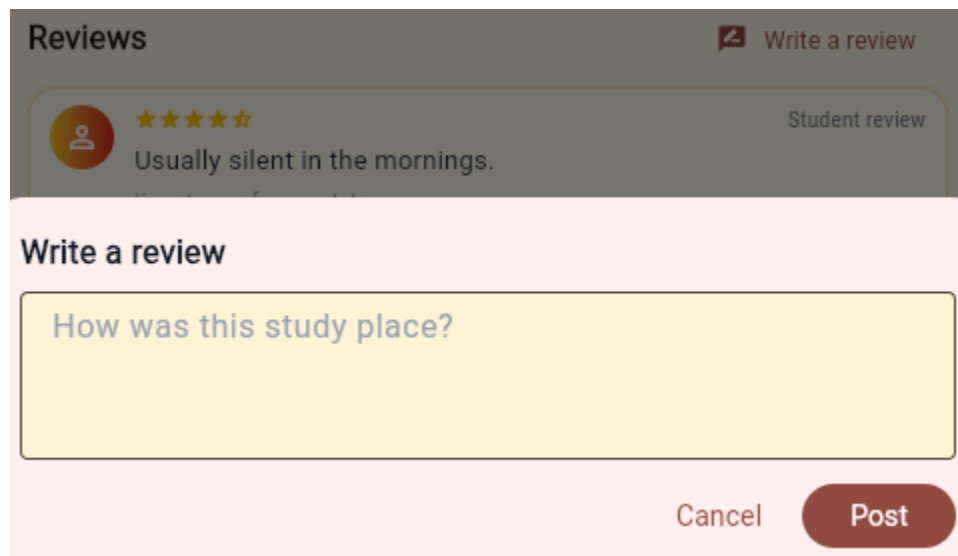
Feedback means the app clearly shows what happened after a tap or change. In DeskDash, tapping the heart on a study card fills it red and the place appears in the Favorites tab. Mark toggles the label

on the details screen and shows a “Visited” badge on the card. Posting a review closes the sheet and adds the new review at the top of the list. When filters or search text change, the list updates immediately, so students see right away that the results have been narrowed. Even in the bus and food panels, the fixed labels like “Buses in next hour” and “Food nearby” act as feedback: they reassure users that these lists are time- and location-based (even though the data is mocked in the prototype).



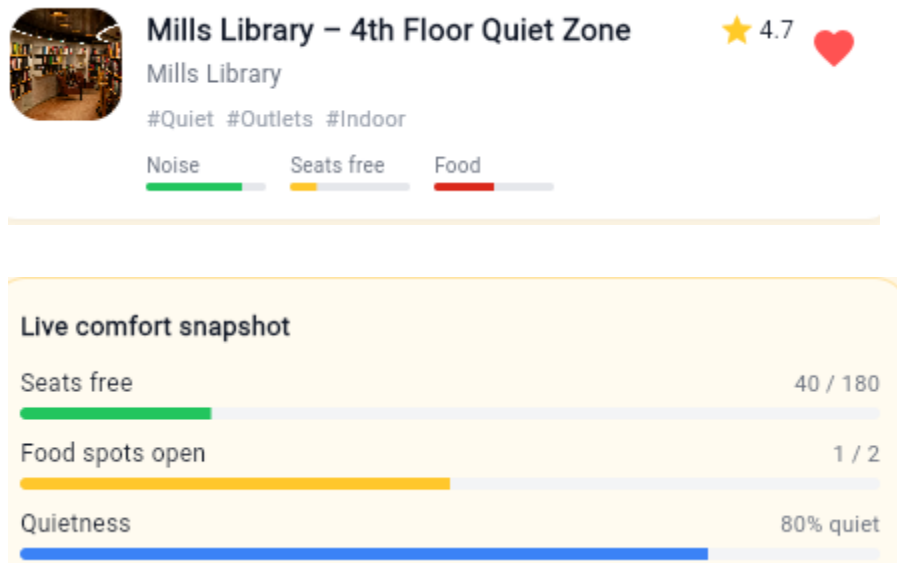
4.3 Constraints

Constraints are limits in the design that guide users toward the right actions and prevent mistakes. In DeskDash, the Write a review sheet only has one text box and two buttons (Cancel and Post), and the Food nearby panel is small and capped in height, so students stay focused on short, useful reviews and the main task of finding a study spot instead of getting lost in extra options.



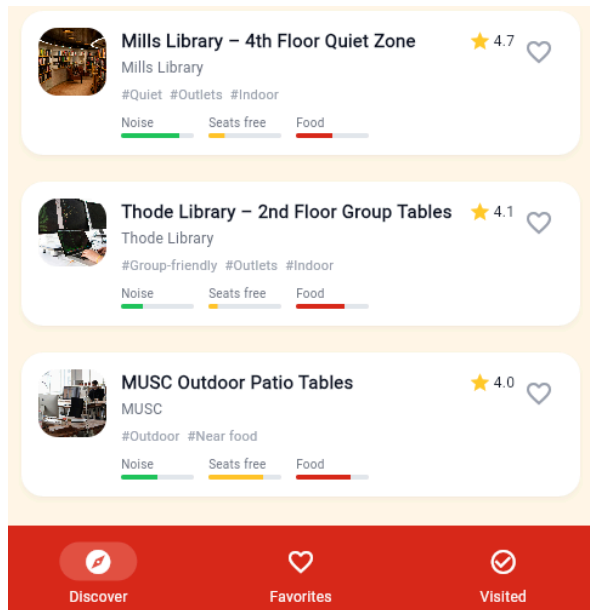
4.4 Mapping

Mapping is how controls line up with what they do, so users can predict the result. In DeskDash, tapping anywhere on a study card always opens that place's details screen with the same photo and name at the top, so the connection is obvious. The Noise / Seats free / Food bars on the cards and in the Live comfort snapshot map higher fill to "better right now" (quieter, more seats free, more food options open). In the Buses in next hour widget, bus icons and route numbers sit on the left, with "in 5 min / in 18 min" on the right, matching how students read transit info. The three bottom tabs map cleanly to the main modes of the app: Discover (all places), Favorites (saved spots), and Visited (places already tried).



4.5 Consistency

Consistency means similar things look and behave in similar ways across the app. In DeskDash, the Discover, Favorites, and Visited tabs all use the same style of study cards (image, title, building, rating, tags, mini bars), and the Save to favorites and Mark visited buttons appear in the same place on every details screen, so once a student learns one screen, the others feel familiar.



4.6 Affordance

Affordance is how the look of something suggests how to use it. In DeskDash, the rounded search bar with a magnifying glass clearly looks like a field you can tap and type into. The pill-shaped filter chips look like toggles you can tap on and off. Study places, food items, and bus trips are all shown as raised cards or rows with shadows, which suggests they are tappable. The large filled Save to favorites button and outlined Mark visited button look like main actions on the details screen. The bottom navigation bar with three labeled icons looks like the standard way to switch sections in a mobile app. All of these visual cues help students work out how to use DeskDash quickly, even without a tutorial.

